NCERT

(Examples)

Botany

Join This Whatsaap Group

Tarun Sir

Living World

1. Family of mango ______________________
2. Contribution of linnaeus ______________________
3. Solanum and petunia family ______________________
4. Felis and Panthera family ______________________
5. Wheat and maize family ______________________
6. Order of mango ______________________
7. Order of solanaceae ______________________
8. Kingdom to genus common character ______________________
9. Order onwards ______________________

Biological Classification

1. Two Kingdom classification ______________________
2. Five Kingdom Classification ______________________
3. Basis of Five Kingdom Classification ______________________

KINGDOM - MONERA

1. Plant bacterial disease ______________________
2. Saprophytic bacteria ______________________
3. Archaebacteria and eubacteria difference ______________________
4. Four example of cyanobacteria ______________________
5. Mycoplasma size ______________________
6. Most metabolic diversity in kingdom ______________________
7. Habitat of methanogen ______________________
8. Bacteria without cell wall ______________________
9. Example of thermoacidophile ______________________

Protista

1. Silica deposit in cell wall of ______________________
2. Golden algae is ______________________
3. Red tide by ______________________
4. Pellicle present in ______________________
5. Chl a and chl b present in ______________________
6. Habitat of Dinoflagellates ______________________
7. Chief producer in ocean ______________________
8. Stiff cellulose plates ______________________
9. True cell wall in slime mould ______________________
10. Flagellated; Protozoan ______________________
11. Ciliated Protozoan ______________________
12. Sporozoans ______________________
13. Amoeboid protozoans ______________________
14. Gullet ______________________

KINGDOM - FUNGI

1. Basis of classification of fungi ______________________
2. Wheat rust ______________________
3. Asexual spores ______________________
4. Sexual spores ______________________
5. Phycomycetes ______________________
6. Antibiotic ______________________

7. Multicellular; Ascomycetes ______________________
8. Unicellular, Ascomycetes ______________________
9. Ascomycetes ______________________
10. Basidiomycetes ______________________
11. Asexual spore absent ______________________
12. Sexual spore absent ______________________
13. Deuteromycetes ______________________

KINGDOM-PLANTAE

1. Insectivorous plant ______________________
2. Parasite ______________________
3. Viral disease ______________________
4. Similar in the size to virus ______________________
5. Disease caused by prions ______________________
6. Disease by Viroids ______________________
7. Contagum vivum fluidum ______________________
8. Crystallise Virus ______________________
9. Genetic material of Bacteriophage ______________________
10. Genetic material of Plant virus ______________________
11. Genetic material not in animal virus ______________________

Plant Kingdom

Algae

1. Isogamous Motile ______________________
2. Isogamous non-motile ______________________
3. Anisogamous ______________________
4. Oogamous ______________________
5. Chlorophyceae ______________________
6. Phaeophyceae ______________________
7. Rhodophyceae ______________________
8. Basis of division of 3 classes of algae ______________________
9. Unequal lateral flagella ______________________
10. Flagella absent ______________________
11. Floridean starch ______________________
12. Pyrenoid ______________________
13. Algae by space traveller ______________________
14. Agar by ______________________
15. Marine algae as food ______________________
16. Diplontic ______________________
17. Haplodiplontic ______________________

Bryophyta

1. Moss ______________________
2. Liverwort ______________________
3. Group ecologically more, economically less Important ______________________
4. Peat moss ______________________
5. Gemmae cup ______________________
6. Multicellular rhizoids ______________________
7. Protonema present ______________________

Pteridophytes

1. First successful land plant ______________________
2. Pteridophytes with cone and rhizome ______________________
3. Pteridophytes (Strobiti or cones) ______________________

4. Pteridophytes (Microphylls) ______________________
5. Pteridophytes (Macrophylls) ______________________
6. Heterosporous ______________________
7. Homosporous ______________________
8. Lycopsida ______________________
9. Sphenopsida ______________________
10. Pteropsida ______________________

Gymnosperm

1. Tallest tree ______________________
2. Branched ______________________
3. Unbranched ______________________
4. Mycorrhiza ______________________
5. Coralloid root ______________________
6. Monoecious ______________________
7. Dioecious ______________________
8. Male and female cone ______________________
9. Only male cone ______________________

Angiosperm

1. Smallest plant ______________________
2. Tallest plant ______________________
3. Trimerous ______________________
4. Pentamerous ______________________

Morphology of Flowering Plant

1. Primary root ______________________
2. Fibrous root ______________________
3. Tap root ______________________
4. Adventitious root ______________________
5. Reticulate Venation ______________________
6. Parallel Venation ______________________
7. Pinnately Compound Leaf ______________________
8. Palmately Compound Leaf ______________________
9. Alternate Phyllotaxy ______________________
10. Opposite Phyllotaxy ______________________
11. Whorled Phyllotaxy ______________________
12. Alternate phylloataxy ______________________
13. Opposite phyllotaxy ______________________
14. Whorled phyllotaxy ______________________
15. Actinomorphic ______________________
16. Zygomorphic ______________________
17. Racemose ______________________
18. Cymose ______________________
19. Asymmetrical ______________________
20. Hypogynous flower(Superior ovary) ______________________
21. Perigynous (half inferior ovary) ______________________
22. Epigynous (Inferior ovary) ______________________
23. Valvate ______________________
24. Twisted ______________________
25. Imbricate ______________________
26. Vexillary ______________________
27. Epipetalous ______________________

28. Epiphyllous ______________________________
29. Monoadelphous - ______________________________
30. diadelphous – ______________________________
31. Polyadelphous - ______________________________
32. Staminode ______________________________
33. Variation in length of filament ______________________________
34. Apocarpous ______________________________
35. Syncarpous ______________________________
36. Axile placentation ______________________________
37. Parietal Placentation ______________________________
38. Marginal Placentation ______________________________
39. Free Central Placentation ______________________________
40. Basal Placentation ______________________________

Family:

1. Malvaceae ______________________________
2. Cruciferae ______________________________
3. Compositae ______________________________
4. Poaceae ______________________________
5. Leguminasae ______________________________

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6. Endospermic seeds ______________________________
7. Non-endospermic seeds ______________________________
8. Character of monocot seeds ______________________________
9. Drupe fruit ______________________________

Anatomy of Flowering Plant

1. Guard Cell (dumb-bell shaped) ______________________________
2. Bean shaped ______________________________
3. Stomatal Apparatus ______________________________
4. Ground Tissue ______________________________
5. Radial Bundle ______________________________s
6. Closed Bundle ______________________________
7. Open ______________________________
8. Casparian strip ______________________________
9. Pericycle, vascular bundle, pith ______________________________
10. Ring vascular bundle ______________________________
11. Scattered Vascular bundle ______________________________
12. Adaxial epidermal cell + vein ______________________________
13. Exarch ______________________________
14. Endarch - ______________________________
15. Bulliform cell ______________________________
16. Starch sheath ______________________________
17. Collenchyma ______________________________
18. Semilunar patch ______________________________
19. Cuticle ______________________________
20. Stomata in epidermal cell ______________________________
21. Layer give root branch ______________________________

Cell- the Unit of Life

1. Smallest Cell ______
2. Largest Isolated single cell ______
3. Inclusion body ______
4. Gas Vacuoles ______
5. Active Transport ______
6. Without membrane organelle in Prokaryotes and Eukaryotes- ______
7. Without membrane in Animal cell ______
8. Not endomembrane system ______
9. RBC protein percentage-membrane ______
10. Amyloplast ______
11. Elaioplast ______
12. aleuroplast ______
13. Chloroplast ______
14. Cell organelle with double membrane ______
15. Cell organelle with single membrane ______
16. Cell organelle with no membrane ______
17. 70s ribosome ______
18. 80s ribosome ______
19. Endomembrane system ______
20. Cisternae, vesicle, tubule ______
21. Classification of chromosome on basis of shape ______
22. Cell wall in algae ______
23. Carotenoid present ______

Biomolecules

1. Acidic Amino acid ______
2. Basic amino acid ______
3. Neutral ______
4. 16 Carbon including carboxyl Carbone ______
5. 20 Carbon Including Carboxy carbone ______
6. Low melting point Oil ______
7. Nucleosides ______
8. Nucleotides ______
9. Polymeric polysaccharide ______
10. Complex Polysaccharide ______
11. Pigments ______
12. Alkaloids Morphine, Codei ______
13. Terpenoids ______
14. Essential oils ______
15. Toxins ______
16. Lectins ______
17. Drugs ______
18. Polymeric subs. ______
19. Element very little in earth crust, 3.3% in human body ______
20. Element 27.7% in eath crust, negligible in human body ______
21. Element present in both in high percentage (46.6% in earth crust and 65% in human body) ______
22. 2 Ring ______)
23. 1 Ring ______
24. Average composition of cell water ______
25. Average composition of cell protein ______

26. Average composition of cell ions ____________________
27. Enable glucose transport into the cell ____________________
28. Starch gives blue colour, cellulose does not due to helical structure in starch Peptide bond ____________
29. Glycosidic bond ____________________
30. Phosphodiester bond ____________________
31. ATP ____________________
32. ADP ____________________
33. Competitive inhibitor ____________________
34. Prosthetic group ____________________
35. Coenzyme ____________________
36. Metal ion ____________________
37. Cellulose present in ____________________
38. Enzyme which introduce double bond ____________________
39. Inhibitor of succinate ____________________
40. Most abundant protein ____________________
41. Quatenary structure ____________________
42. Shape of graph for temp and enzyme ____________________
43. Structural polysaccharide ____________________
44. Reducing sugar ____________________
45. Diasaccharide ____________________
46. S-containing amino acid ____________________
47. OH- Group containing ____________________
48. Most complex amino acid ____________________

Cell Cycle & Cell Division

1. Duration of M-phase ____________________
2. Active mitosis ____________________
3. Active meiosis ____________________
4. Compaction of chromosome ____________________
5. Synapsis (bivalent/tetrad), Synaptonemal complex ____________________
6. Crossing over, clear tetrad ____________________
7. Dissolution of synaptonemal complex Diplotene ____________________
8. Terminalisation of chiasmata ____________________
9. Sister chromatid separate ____________________
10. Homologus pair separate ____________________
11. Most condensed chromosome-mitosis ____________________
12. Cell plate method ____________________
13. Amount of DNA 4c ____________________
14. Amount of DNA 2c ____________________
15. Mitosis in haploid animal cell ____________________
16. Centriole double after meiosis I ____________________
17. Centriole double in ____________________
18. Recombinase active ____________________

Photosynthesis

1. Priestley : ____________________
2. Jan Ingenhousz : ____________________
3. Julius von sachs : ____________________
4. T.W. engelmann : ____________________
5. Light reaction : ____________________

6. PS-I : ____________________
7. PS-II : ____________________
8. Cyclic photophosphorylation : ____________________
9. Granal thylakoid ____________________
10. Stromal lamellae ____________________
11. C3 -Plant : ____________________
12. C4-Plant : ____________________
13. First product of Photorespiration ____________________
14. Sugar in C-3 Cycle ____________________
15. First product of C-4 cycle ____________________
16. Hydrogen Carrier in non cyclic ____________________
17. Enzyme for NADPH ____________________
18. Green house plant ____________________
19. Saturation of C-4 ____________________
20. Internal CO2 conc. ____________________
21. Reaction centre- ____________________
22. Antennae ____________________
23. Chl a colour ____________________
24. Chl b colour ____________________
25. Xanthophyll colour ____________________

Respiration

1. Common pathway for eukaryote and prokaryote : ____________________s
2. Acetyl CoA : ____________________
3. RQ for fat ____________________
4. First enzyme of glycolysis ____________________
5. Sucrose breakdown ____________________
6. Respiratory substrate ____________________
7. Last enzyme of glycolysis ____________________
8. First enzyme of kreb cycle ____________________
9. Succintate to fumarate ____________________
10. Pyruvate to acetyl coA ____________________
11. Complex accept from NADH ____________________
12. Complex IV ____________________
13. Complex II ____________________
14. Complex V ____________________
15. Lactic acid ____________________
16. Acetaldehyde from pyruvate ____________________
17. Ethanol ____________________

Plant Growth & Development

1. Indole compound : ____________________
2. Adenine derivative : ____________________
3. Carotenoids : ____________________
4. Terpenes : ____________________
5. Gases : ____________________
6. Growth Promoters : ____________________
7. Growth Inhibitor : ____________________
8. Largely inhibitor : ____________________
9. Natural auxin : ____________________
10. Synthetic auxin : ____________________

11. Rapid internode / petiole elongation in deep water rice plants: ____________________
12. Kill dicot weed : ____________________
13. From autoclaved herring sperm DNA : ____________________
14. From corn kernels and coconut milk : ____________________
15. Initiate flowering and synchronising fruit set in pineapple: ____________________
16. Initiate germination in peanut seeds, sprouting of potato tuber : ____________________
17. Differentiation : ____________________
18. Dedifferentiation : ____________________
19. Redifferentiation : ____________________
20. Plasticity : ____________________
21. Bakane (disease of rice seedling) ____________________
22. First isolated from human urine ____________________
23. Auxin promote flowering ____________________
24. Auxins induce parthenocarpy ____________________
25. Ethephon hasten fruit ripening ____________________
26. Accelerate absession in ____________________

Sexual Reproduction in Flowering Plant

1. Pollen allergy ____________________
2. Long viability of pollen ____________________
3. One ovule in ovary ____________________
4. Many ovule in ovary ____________________
5. Water pollinated plant ____________________
6. Aquatic plant pollinate by wind and water ____________________
7. Pollinator ____________________
8. Tallest flower ____________________
9. Monoecious (Male & female flower on same plant)____________________
10. Non-albuminous seed ____________________
11. Albuminous seed ____________________
12. False fruit ____________________
13. Ribbon shape pollen ____________________
14. Mono ovular ovary ____________________
15. Cleistogamous and chasmogamous flower ____________________
16. Pollination on surface of water ____________________
17. Foul smell flower pollination by ____________________
18. Yucca plant pollination ____________________
19. Scutellum ____________________
20. Apomixix-nucellus ____________________
21. Polyembryony ____________________
22. Long viability seed-2000 yr ____________________
23. Long viability of seed – 10000yr ____________________
24. Perisperm ____________________

Principle of Inheritances & Variation

1. Incomplete dominance ____________________
2. X O type of sex determination ____________________
3. X Y chromosome ____________________
4. Co-dominance ____________________
5. Pleiotorpy ____________________
6. Polygenic inheritance ____________________

7. Experimental verification of chromosomal theory ______
8. Multiple allele ______
9. 1.3 percent recombination ______
10. X-body by ______
11. Thallasemia chromosome ______
12. Trisomy of 21st chromosome ______
13. Linkage map ______
14. Cross with recessive parent ______
15. Mendelian disorder ______
16. Chromosomal disorder ______
17. Haplodiploidy ______
18. Sex-linked recessive ______

Molecular Basis of Inheritance

1. Purine ______
2. Pyrimidines ______
3. Nuclein ______
4. X-ray diffraction ______
5. Central dogma ______
6. Base pair in nucleosome ______
7. Thymine name ______
8. Most condensed chromatin ______
9. Transforming principle in Griffith ______
10. Transforming principle in Avery,mccarty,macleod ______
11. Unequivocal proof of genetic material ______
12. Harshey and chase use ______
13. Radioactivity when P^{32} use ______
14. Taylor use ______
15. Messelson and stahl centrifugation ______
16. Open DNA in Replication ______
17. Start codon ______
18. Stop codon ______
19. First decipher codon ______
20. tRNA of 30s subunit ______
21. RNA poly III ______
22. Part of mRNA not translated ______
23. Ribozyme ______
24. Split gene ______
25. Regulatory gene ______
26. Structural gene of lac operon ______
27. Inducer of lac operon lac ______
28. Protein for initiation of transcription ______
29. Gene cloning in HGP ______
30. Satellite in DNA fingerprinting ______

MICROBES IN HUMAN WELFARE

1. Baker’s yeast ______
2. Produce without distillation ______
3. Produce by distillation ______
4. First antibiotic ______

5. Acid Producer ______
6. Detergent formulation ______
7. Microbial bicontrol agent ______
8. fix atmosphric nitrogen (free-living) ______
9. Dragon fly kill ______
10. Ladybird ______
11. LAB produce ______
12. Establish role of antibiotic ______
13. Immunosuppressive agent ______
14. Cholesterol lowering ______
15. Mass of bacteria and fungi in aeration tank ______
16. Gas in STP ______
17. Biogas plant bacteria ______
18. Genus of mycorrhizae ______
19. IPM by ______
20. Swiss cheese ______i
21. Baker's yeast ______

BIOTECHNOLOGY : PRINCIPLE & PROCESS

1. Restriction enzyme ______
2. Cloning vector ______
3. Useful Selectable Marker for E-coli ______
4. Vector for cloning gene in plant ______
5. Vector for cloning gene in animal ______
6. Enhanced nutritional value of food ______
7. Protein enended by gene ______
8. Transgenic Animal ______
9. Human protein used to treat emphysema ______
10. First transgeneic cow ______

ORGANISM & POPULATION

1. Breed only once in their lifetime ______
2. Breed many times ______
3. Produce large number of small sized offspring ______
4. Produce small size of number of large sized offspring ______
5. Introduced into Australia in early 1920 ______
6. Predator ______
7. extinct within a decade after goats were introduced on the island ______
8. Lice on human & ticks on dog ,Cuscuta, Parasitic plant on hidge plant ______
9. Various pathogenic bacteria ______
10. Virus, fungi, protozoan, flat worm, round worm Brood Parasitism in birds ______
11. Growing on back of whale ______
12. Growing as an epiphyte on Mango branch ______
13. Close association of ______
14. Macarthur warbelers ______
15. Barnacle -gause exclusion ______
16. Flamingo and fish compete for ______
17. Pre reproductive and reproductive equal ______
18. Pug mark and faecal pellet ______

Ecosystem

1. Man made ecosystem ______
2. Vertical distribution of species ______
3. Unit of productivity ______
4. Available biomass for next trophic level ______
5. Detritivore rich in lignin and chitin ______
6. Rich in sugar and nitrogen ______
7. 50 percent of total light ______
8. Inverted pyramid ______
9. Aquatic more food chain ______
10. Land more food chain ______
11. Trophic level of secondary carnivore ______
12. Black amorphous, colloidal ______
13. Biomass of living at trophic level ______

Biodiversity and conservation

1. Robert may ______
2. Genetic level biodiversity ______
3. Equator more diversity in ______
4. India area and biodiversity ______
5. Maximum biodiversity ______
6. Species area relation ship ______
7. Long term External plot experiment ______
8. Extinct from Russia and Africa ______
9. Value of Z ______
10. Niche specialization ______
11. IUCN threatened -most ______
12. Broadly utilitarian ______
13. Lake Victoria-invasion by ______
14. Over exploitation ______
15. Habitat loss ______
16. Hot spot ______
17. Meghalaya-sacred groove ______
18. Wing in rivet popper ______
19. Ex situ ______
20. Earth summit ______
21. Ramasar site ______
22. African cat fish ______

Answer Key

Living World

1. Family of mango — Anacardiaceae
2. Contribution of linnaeus - — Binomial nomenclature , Systemna naturae
3. Solanum and petunia family – — Solanaceae
4. Felis and Panthera family — Felidae
5. Wheat and maize family — Poaceae
6. Order of mango — Sapindales
7. Order of solanaceae — Polymoniales
8. Kingdom to genus common character — Increase
9. Order onwards — Aggregation of trait

Biological Classification

1. Two Kingdom classification — Linneaus
2. Five Kingdom Classification — R.H. Whittaker. 1969
3. Basis of Five Kingdom Classification — Cell Structure, Body organisation, Mode of Nutrition, Reproduction, Phylogenetic relationship

KINGDOM - MONERA

1. Plant bacterial disease — Citrus canker
2. Saprophytic bacteria — Antibiotic producing ,curd making, N2 fixation
3. Archaebacteria and eubacteria difference — Cell wall
4. Four example of cyanobacteria — nostoc ,Anabena, Oscillatoria, Spirulina
5. Mycoplasma size — 0.3 micrometer
6. Most metabolic diversity in kingdom — Monera
7. Habitat of methanogen — Gut of ruminants, marshy area
8. Bacteria without cell wall — Mycoplasma
9. Example of thermoacidophile — Thermus aquaticus

Protista

1. Silica deposit in cell wall of — Diatom
2. Golden algae is — Desmids
3. Red tide by — Gonylaux
4. Pellicle present in — Euglenoids
5. Chl a and chl b present in — Euglenoids
6. Habitat of Dinoflagellates — Marine
7. Chief producer in ocean — Diatom
8. Stiff cellulose plates — Dinoflagellates
9. True cell wall in slime mould — Spore stage
10. Flagellated; Protozoan — Trypanosoma
11. Ciliated Protozoan — Paramoecium
12. Sporozoans — Plasmodium
13. Amoeboid protozoans — Entamoeba
14. Gullet — Paramoecium

KINGDOM - FUNGI

1. Basis of classification of fungi — Morphology and mycelium, mode of spore formation, fruiting body
2. Wheat rust — Puccinia
3. Asexual spores — Conidia, sporangiospores, zoospores
4. Sexual spores — Oospores, ascospores, basidiospores, zygospore

5.	Phycomycetes	Mucor, Rhizopus (bread mould), Albugo (parasitic fungi on mustard)
6.	Antibiotic by ascomycetes	Penicillium
7.	Multicellular; Ascomycetes	Penicillum
8.	Unicellular, Ascomycetes	Yeast (Saccharomyces)
9.	Ascomycetes	Aspergillus, Claviceps, Neurospora, Morels, Truffles, Penecillium, Yeast
10.	Basidiomycetes	Agaricus (Mushroom), Ustilago (Smut), Pucinia(wheat rust), Puffballs, Bracket, fungi
11.	Asexual spore absent	Basidiomycetes
12.	Sexual spore absent	Deuteromycetes
13.	Deuteromycetes	Alternaria, Colletotricum,Trichoderma

KINGDOM-PLANTAE

1.	Insectivorous plant	Bladderwort, Venus fly Trap
2.	Parasite	Cuscuta
3.	Viral disease	Mumps, small pox, Herpes, Influenza
4.	Similar in the size to virus	Prions
5.	Disease caused by prions	Bovine spongiform encephalopathy (BSE) called mad cow disease in cattle , Cr- jacob disease in humans
6.	Disease by Viroids	T.O diener
7.	Contagum vivum fluidum	Beijerinek
8.	Crystallise Virus	Stanley
9.	Genetic material of Bacteriophage	dsDNA
10.	Genetic material of Plant virus	ssRNA
11.	Genetic material not in animal virus	ssDNA

Plant Kingdom

Algae

1.	Isogamous Motile	Ulothrix ,Cladophora
2.	Isogamous non-motile	Spirogyra
3.	Anisogamous	Eudorina
4.	Oogamous	Volvox, fucus
5.	Chlorophyceae	Chlamydomonas, Volvox, Ulothrix, Spirogyra, Chara
6.	Phaeophyceae	Ectocorpus, Dictyota, Laminaria, Sargassum ,fucus
7.	Rhodophyceae	Polysiphonia, Porphyra, Gracilaria, Gelidium
8.	Basis of division of 3 classes of algae	Type of pigment possessed and type of stored food
9.	Unequal lateral flagella	Brown algae
10.	Flagella absent	Red algae
11.	Floridean starch	Red algae
12.	Pyrenoid	Green algae
13.	Algae by space traveller	Chlorella
14.	Agar by	Red algae
15.	Marine algae as food	Sargassum,laminaria,Porphyra
16.	Diplontic	Fucus
17.	Haplodiplontic	Kelp ,Ectocarpus, Polysiphonia

Bryophyta

1.	Moss	Sphagnum, funaria, Polytrichum
2.	Liverwort	Marchantia
3.	Group ecologically more, economically less Important	Bryophyte
4.	Peat moss	Sphegnum
5.	Gemmae cup	Marchantia
6.	Multicellular rhizoids	Moss

7.	Protonema present	Moss

Pteridophytes

1.	First successful land plant	Pteridophytes
2.	Pteridophytes with cone and rhizome	Equistum (horstail)
3.	Pteridophytes (Strobiti or cones)	Selaginella, Equisetum
4.	Pteridophytes (Microphylls)	Selaginella
5.	Pteridophytes (Macrophylls)	Ferns
6.	Heterosporous	Selaginella and Salvinia
7.	Homosporous	Dryopteris, Pteris
8.	Lycopsida	lycopodium and seleginella
9.	Sphenopsida	Equisetum
10.	Pteropsida	Pteris,Dryopteris,Adiantum ,salvinia

Gymnosperm

1.	Tallest tree	Sequoia
2.	Branched	Pinus cedrus
3.	Unbranched	Cycas
4.	Mycorrhiza	Pinus
5.	Coralloid root	Cycas
6.	Monoecious	Pinus
7.	Dioecious	Cycas
8.	Male and female cone	Pinus
9.	Only male cone	Cycas

Angiosperm

1.	Smallest plant	Wolffia
2.	Tallest plant	Eucalyptus
3.	Trimerous	Monocot
4.	Pentamerous	Dicot

Morphology of Flowering Plant

1.	Primary root	Direct elongation of radicle
2.	Fibrous root	Wheat
3.	Tap root	Mustard
4.	Adventitious root	Monstera, grass, banyan tree
5.	Reticulate Venation	Dicotyledonous
6.	Parallel Venation	Monocotyledons
7.	Pinnately Compound Leaf	Neem
8.	Palmately Compound Leaf	Silk cotton
9.	Alternate Phyllotaxy	China Rose, Mustard, Sunflower
10.	Opposite Phyllotaxy	Calotropis, Guava plant
11.	Whorled Phyllotaxy	Alstonia
12.	Alternate phylloataxy	Mustard,sunflower,china rose
13.	Opposite phyllotaxy	Guava, Calotropis
14.	Whorled phyllotaxy	Alstonia
15.	Actinomorphic	Mustard, chilli, datura
16.	Zygomorphic	Pea, bean, gulmohar, Cassia and fabaceae
17.	Racemose	Brassicaceae ,fabaceae ,poaceae ,asterceae
18.	Cymose	Solanaceae , Liliaceae ,malvaceae
19.	Asymmetrical	Canno

20.	Hypogynous flower(Superior ovary)	Mustard, China rose, brinjal
21.	Perigynous (half inferior ovary)	Plum, Rose, Peach
22.	Epigynous (Inferior ovary)	Guava, Cucumber, ray floret of sunflower
23.	Valvate	Calotropis
24.	Twisted	China rose, Lady finger, Cotton
25.	Imbricate	Cassia, gulmohar
26.	Vexillary	Pea, bean
27.	Epipetalous	Brinjal
28.	Epiphyllous	Lily
29.	Monoadelphous -	China rose
30.	diadelphous –	Pea
31.	Polyadelphous -	Citrus
32.	Staminode	Sterile stamen
33.	Variation in length of filament	Salvia, Mustard
34.	Apocarpous	Lotus, Rose, Michellia
35.	Syncarpous	Mustard, Tomato, papaver, hibiscus
36.	Axile placentation	China rose, tomato, lemon
37.	Parietal Placentation	Mustard, Argemone
38.	Marginal Placentation	Pea
39.	Free Central Placentation	Dianthus, Primrose
40.	Basal Placentation	Sunflower, Marigold

Family:

1.	Malvaceae	Hibiscus (China rose), Bhindi (Okra), Cotton
2.	Cruciferae	Mustard, Cabbage, Raddish, turnip, Cauliflower
3.	Compositae	Sunflower, Marigold
4.	Poaceae	Wheat,Maize,rice,Oat,Barley ,sugarcane ,Grass, Bamboo .
5.	Leguminasae	Source of pulse Gram, arhar, sem, moong, Soybean, Dye - Indigofera, fibre - sunbean, fodder – sesbania, Trifolium, Ornament - Lupin, Sweet Pea, Medicine - Mulaithi
6.	Endospermic seeds	Monocot (exception - orchid)
7.	Non-endospermic seeds	Bean, Gram, Pea (most dicots, except castor)
8.	Character of monocot seeds	Coleoptile, Coleorhiza, scutellum, epiblast)
9.	Drupe fruit	Mango, Coconut- hard endocarp and single seed

Anatomy of Flowering Plant

1.	Guard Cell (dumb-bell shaped)	Grasses (monocots)
2.	Bean shaped	Dicots
3.	Stomatal Apparatus	Stomatal Aperture, Guard cell, Subsidiary cell
4.	Ground Tissue	All tissue except epidermis and vascular tissue
5.	Radial Bundle	Roots
6.	Closed Bundle	Monocot root, stem and leaf and dicot leaf and root
7.	Open	Dicot Stem
8.	Casparian strip	Waxy material (Suberin) in endodermis
9.	Pericycle, vascular bundle, pith	Stele
10.	Ring vascular bundle	Dicot stem
11.	Scattered Vascular bundle	Monocot stem
12.	Adaxial epidermal cell + vein	Bulliform cell
13.	Exarch	Protoxylem outer- Root
14.	Endarch -	Protoxylem inner- Stem

15.	Bulliform cell	Adaxial on grass leaf
16.	Starch sheath	Endoermis dicot stem
17.	Collenchyma	Hypodermis dicot stem
18.	Semilunar patch	Pericycle of dicot stem
19.	Cuticle	leaf and stem
20.	Stomata in epidermal cell	Guard cell
21.	Layer give root branch	pericycle in dicot stem

Cell, the Unit of Life

1.	Smallest Cell	Mycoplasma
2.	Largest Isolated single cell	egg. of ostrich
3.	Inclusion body	Phosphate, granules, Cyanophycean granule, glycogen Granules and gas vacuole
4.	Gas Vacuoles	Blue green & purple and green photosynthetic bacteria
5.	Active Transport	Na+/K+ pump
6.	Without membrane organelle in Prokaryotes and Eukaryotes- Ribosome	
7.	Without membrane in Animal cell	Centriole
8.	Not endomembrane system	Peroxisome, Mitochondria, Chloroplast
9.	RBC protein percentage-membrane	52 percent
10.	Amyloplast	Store Carbohydrate (Potato)
11.	Elaioplast	Store oil & fats
12.	aleuroplast	Store protein
13.	Chloroplast	Double Membrane
14.	Cell organelle with double membrane	Mitochondria, chloroplast, nucleus
15.	Cell organelle with single membrane	Endoplasmic reticulum, golgi body, vacuole, lysosome, Microbodies
16.	Cell organelle with no membrane	Ribosome, nucleolus, centriole
17.	70s ribosome	Both prokaryote and eukaryote
18.	80s ribosome	Only eukaryote
19.	Endomembrane system	ER, Golgi body, vacuole, Lysosome
20.	Cisternae, vesicle, tubule	Golgi body ER
21.	Classification of chromosome on basis of shape	Metacentric (V), Submetacentric (L), Acrocentric (J), Telocentric (I)
22.	Cell wall in algae	Cellulose galactans mannas and minerals like calcium Carbonate
23.	Carotenoid present	Chloroplast and chromoplast

Biomolecules

1.	Acidic Amino acid	Glutamic acid
2.	Basic amino acid	Lysine, Arginine
3.	Neutral	Valine
4.	16 Carbon including carboxyl Carbone	Palmitic acid
5.	20 Carbon Including Carboxy carbone	Arachidonic acid
6.	Low melting point Oil	Gingelly oil
7.	Nucleosides	Adenosine guomosine cytidine uridic thymidine
8.	Nucleotides	Adenylic acid, thymidylic acid , guanylic acid, Uridylic acid, Cytidylic acid
9.	Polymeric polysaccharide	Cellulose , starch
10.	Complex Polysaccharide	Chitin
11.	Pigments	Carotenoids, Anthocyanins
12.	Alkaloids	Morphine, Codeine

13.	Terpenoids	Monoterpenes, Diterpenes
14.	Essential oils	Lemon grass oil
15.	Toxins	Abrin, Ricin
16.	Lectins	Concanavalin A
17.	Drugs	Vinblastin, curcumin
18.	Polymeric subs.	Rubber, gums, cellulose
19.	Element very little in earth crust, 3.3% in human body	Nitrogen
20.	Element 27.7% in eath crust, negligible in human body	Silicon
21.	Element present in both in high percentage (46.6% in earth crust and 65% in human body)	Oxygen
22.	2 Ring	Purines (A, G)
23.	1 Ring	Pyrimidine (C, T, U)
24.	Average composition of cell water	70-90%
25.	Average composition of cell protein	10-15%
26.	Average composition of cell ions	1%
27.	Enable glucose transport into the cell	GLUT-IV
28.	Starch gives blue colour, cellulose does not due to helical structure in starch Peptide bond	Protein
29.	Glycosidic bond	Carbohydrate
30.	Phosphodiester bond	Nucleic acid
31.	ATP	2 high energy bond.
32.	ADP	1 high energy bond
33.	Competitive inhibitor	Malonate
34.	Prosthetic group	Peroxidase catalase
35.	Coenzyme	NAD, NADP
36.	Metal ion	Zn, Carboxypeptidase
37.	Cellulose present in	Cotton
38.	Enzyme which introduce double bond	Lyase
39.	Inhibitor of succinate	Malonate
40.	Most abundant protein	Rubisco
41.	Quatenary structure	Haemoglobin
42.	Shape of graph for temp and enzyme	Bell shape
43.	Structural polysaccharide	Chitin, cellulose
44.	Reducing sugar	Sucrose
45.	Diasaccharide	Maltose,Sucrose ,lactose
46.	S-containing amino acid	Methionine,cystiene
47.	OH- Group containing	Seriene,Tyrosine
48.	Most complex amino acid	Tryptophan

Cell Cycle & Cell Division

1.	Duration of M-phase	1 hour
2.	Active mitosis	root tip and shoot tip
3.	Active meiosis	Megaspore mother cell, microspore mother cell
4.	Compaction of chromosome	Leptotene
5.	Synapsis (bivalent/tetrad), Synaptonemal complex	Zygotene
6.	Crossing over, clear tetrad	Pachytene
7.	Dissolution of synaptonemal complex	Diplotene
8.	Terminalisation of chiasmata	Diakinesis
9.	Sister chromatid separate	Anaphase ,Anaphase II
10.	Homologus pair separate	Anaphase I
11.	Most condensed chromosome-mitosis	Metaphase
12.	Cell plate method	Plant
13.	Amount of DNA 4c	G2

14.	Amount of DNA 2c	G1
15.	Mitosis in haploid animal cell	Drone
16.	Centriole double after meiosis I	interkinesis
17.	Centriole double in	S-phase
18.	Recombinase active	Pachytene

Photosynthesis

1.	Priestley :	Discovered - Oxygen [Bell jar experiment]
2.	Jan Ingenhousz :	Sunlight essential
3.	Julius von sachs :	Glucose
4.	T.W.engelmann :	Cladophora (Green algae) , First action spectrum
5.	Light reaction :	ATP, NADPH production and Oxygen
6.	PS-I :	700 nm
7.	PS-II :	680 nm
8.	Cyclic photophosphorylation :	PS-I only
9.	Granal thylakoid	PS I and PS II
10.	Stromal lamellae	PS-I
11.	C3 -Plant :	Most plant
12.	C4-Plant :	Maize, Sorghum
13.	First product of Photorespiration	Phosphoglycolate
14.	Sugar in C-3 Cycle	triose phosphate ,RUBP
15.	First product of C-4 cycle	OAA
16.	Hydrogen Carrier in non cyclic	Plasto quinone
17.	Enzyme for NADPH	FNR
18.	Green house plant	C-3 plant -bell pepper ,tomato
19.	Saturation of C-4	450 ppm
20.	Internal CO2 conc.	Internal factor
21.	Reaction centre-	Chl a
22.	Antennae	Chl b ,carotenoids ,xanthophyll
23.	Chl a colour	Blue green
24.	Chl b colour	Green yellow
25.	Xanthophyll colour	Yellow

Respiration

1.	Common pathway for eukaryote and prokaryote :	Glycolysis
2.	Acetyl CoA :	Common to fats, carbohydrates, Proteins
3.	RQ for fat	0.7
4.	First enzyme of glycolysis	hexokinase
5.	Sucrose breakdown	invertase
6.	Respiratory substrate	organic acid,carbohydrate,fat,proteins
7.	Last enzyme of glycolysis	Pyruvate kinase
8.	First enzyme of kreb cycle	Citrate synthetase
9.	Succintate to fumarate	Succinate dehydrogenase
10.	Pyruvate to acetyl coA	Pyruvate dehydrogenase
11.	Complex accept from NADH	NADH dehydrogenase
12.	Complex IV	cytochrome c oxidase
13.	Complex II	Succinate dehydrogenase
14.	Complex V	F0-F1 or ATP synthetase
15.	Lactic acid	lactate dehydrogenase
16.	Acetaldehyde from pyruvate	pyruvate decarboxylase
17.	Ethanol	alcohol dehydrogenase

Plant Growth & Development

1. Indole compound :	IAA
2. Adenine derivative :	N6-furfurylamino purine, kinetin
3. Carotenoids :	ABA
4. Terpenes :	Gibberellic acid
5. Gases :	Ethylene
6. Growth Promoters :	Auxin, Gibberellic acid, Cytokinin
7. Growth Inhibitor :	ABA
8. Largely inhibitor :	Ethylene
9. Natural auxin :	IAA, IBA
10. Synthetic auxin :	2, 4-D, NAA
11. Rapid internode / petiole elongation in deep: water rice plants	Ethylene
12. Kill dicot weed :	2, 4-D
13. From autoclaved herring sperm DNA :	Kinetin
14. From corn kernels and coconut milk :	Zeatin
15. Initiate flowering and synchronising fruit: set in pineapple	Ethylene
16. Initiate germination in peanut seeds, sprouting: of potato tuber	Ethylene
17. Differentiation :	Tracheary elements
18. Dedifferentiation :	Interfasicular cambium and cork cambium
19. Redifferentiation :	2^0 xylem and phloem
20. Plasticity :	Heterophylly in cotton, coriander and larkspur
21. Bakane (disease of rice seedling)	Gibberella fujikuroi
22. First isolated from human urine	Auxin
23. Auxin promote flowering	Pineapple
24. Auxins induce parthenocarpy	Tomato
25. Ethephon hasten fruit ripening	Tomato, apple
26. Accelerate absession in	Thinning of cotton, cherry, walnut

Sexual Reproduction in Flowering Plant

1.	Pollen allergy	parthenium
2.	Long viability of pollen	Rosaceae, solanaceae ,Leguminosae
3.	One ovule in ovary	Wheat, Paddy, Mango
4.	Many ovule in ovary	Papaya, Watermelon, Orchids
5.	Water pollinated plant	Vallesneria, Hydrilla
6.	Aquatic plant pollinate by wind and water	Water hyacinth, water lilly
7.	Pollinator	a. Primates (lemur), arboreal (tree-dwelling), rodent, even reptile (gecko lizard), garden lizard
8.	Tallest flower	Amorphophallus (6 feet in height)
9.	Monoecious (Male & female flower on same plant)	Castor, Maize
10.	Non-albuminous seed	Pea, groundnut, beans
11.	Albuminous seed	Wheat, Maize, Barley, Castor
12.	False fruit	Strawberry,cashew,apple
13.	Ribbon shape pollen	Zostera
14.	Mono ovular ovary	wind pollinated
15.	Cleistogamous and chasmogamous flower	oxalis,commelina,viola
16.	Pollination on surface of water	Hydrilla ,Vallisneria

17.	Foul smell flower pollination by	Beetle and flies
18.	Yucca plant pollination	Moth
19.	Scutellum	Grass family seed cotyledon
20.	Apomixix-nucellus	Mango
21.	Polyembryony	Orange
22.	Long viability seed-2000 yr	date palm-phoenix dactylifera
23.	Long viability of seed – 10000yr	Lupinus arcticus
24.	Perisperm	Beet and black pepper

Principle of Inheritances & Variation

1.	Incomplete dominance	Dog flower (snapdragon or Antirrhium sp.)
2.	X O type of sex determination	Grasshopper
3.	X Y chromosome	Male (human), drosophila
4.	Co-dominance	ABO blood group
5.	Pleiotorpy	starch synthesis gene, hydroxylase
6.	Polygenic inheritance	skin colour ,human height
7.	Experimental verification of chromosomal theory	T.H morgan
8.	Multiple allele	blood group
9.	1.3 percent recombination	body colour and eye colour
10.	X-body by	Henking
11.	Thallasemia chromosome	chr-11 and 16
12.	Trisomy of 21^{st} chromosome	Down's syndrome
13.	Linkage map	Sturtevant
14.	Cross with recessive parent	test cross
15.	Mendelian disorder	Haemophilia, cystic fibrosis, sickle cell anemia, colour blindness, phenylketonuria, thalassemia
16.	Chromosomal disorder	Down's syndrome, Klinefelter's syndrome Turner's Syndrome.
17.	Haplodiploidy	Honeybee(Queen-2n,worker-2n,Drone-n)
18.	Sex-linked	recessive Haemophilia, colourblindness

Molecular Basis of Inheritance

1.	Purine	Adenine & Guanine
2.	Pyrimidines	Cytosine, Uracil, Thymine
3.	Nuclein	meischer
4.	X-ray diffraction	Wilkins and franklin
5.	Central dogma	Crick
6.	Base pair in nucleosome	200
7.	Thymine name	5-methyl uracil
8.	Most condensed chromatin	heterochromatin
9.	Transforming principle in Griffith	Unknown
10.	Transforming principle in Avery,mccarty,macleod	DNA
11.	Unequivocal proof of genetic material	Harshey and chase
12.	Harshey and chase use	P^{32} and S^{35}
13.	Radioactivity when P^{32} use	Pellet
14.	Taylor use	Radioacitivity thymidine
15.	Messelson and stahl centrifugation	CsCl density centrifugation
16.	Open DNA in Replication	Helicase
17.	Start codon	AUG
18.	Stop codon	UAA,UGA,UAG

19.	First decipher codon	UUU-phe
20.	tRNA of 30s subunit	16s
21.	RNA poly III	tRNA,5s,SnRNA
22.	Part of mRNA not translated	UTR
23.	Ribozyme	23s rRNA
24.	Split gene	Eukaryotes
25.	Regulatory gene	Lac I
26.	Structural gene of lac operon	lac z,y,a
27.	Inducer of lac operon lac	lactose ,allolactose
28.	Protein for initiation of transcription	sigma factor and RNA polymerase
29.	Gene cloning in HGP	YAC and BAC
30.	Satellite in DNA fingerprinting	Minisatellite

MICROBES IN HUMAN WELFARE

1.	Baker's yeast	Yeast (Saccharomyces Cerevisiae)
2.	Produce without distillation	Wine, beer
3.	Produce by distillation	Whisky, brandy, rum
4.	First antibiotic	Penicillin
5.	Acid Producer	Aspergillus niger (a fungus) of citric acid Acetobacter aceti (a bacterium) of acetic acid Clostridium butyticum (a bacterium) of butyric acid Lactobacillus (a bacterium) of Lactic acid Common Production of ethanol Yeast
6.	Detergent formulation	Lipases
7.	Microbial bicontrol agent	bacillus thuringenesis
8.	fix atmosphric nitrogen (free-living)	Azospirillum, Azotobacter
9.	Dragon fly kill	Mosquito
10.	Ladybird	Aphids
11.	LAB produce	vit B12
12.	Establish role of antibiotic	Chain and florey
13.	Immunosuppressive agent	cyclosporin A-trichoderma polysporum
14.	Cholesterol lowering	Statin -Monascus
15.	Mass of bacteria and fungi in aeration tank	Flocs
16.	Gas in STP	H_2S,CH_4 and CO_2
17.	Biogas plant bacteria	Methanogen
18.	Genus of mycorrhizae	Glomus
19.	IPM by	Baculovirus
20.	Swiss cheese	Propiniobacterium sharmanii
21.	Baker's yeast	Sachharomyces

BIOTECHNOLOGY : PRINCIPLE & PROCESS

1.	Restriction enzyme	Hind II, EcorI (come from Escherichia Coli. R113)
2.	Cloning vector	Plasmid, bacteriophage (PBR 322)
3.	Useful Selectable Marker for E-coli	ampicillin, chloramphenicol, tetracycline, Kanamycin
4.	Vector for cloning gene in plant	Agrobacterium Tumifaciens
5.	Vector for cloning gene in animal	Retroviruses
6.	Enhanced nutritional value of food	golden rice i.e., vitamin A enriched rice
7.	Protein enended by gene	CryIAc (Cotton boll worm) & CryIIAb (Cotton boll worm) a. CryIAb (Corn borer)
8.	Transgenic Animal	Transgenic rat, rabbits, Pigs, Sheep, Cow, Fish
9.	Human protein used to treat emphysema	α-1 antitrypsin

10.	First transgeneic cow	Rosie

ORGANISM & POPULATION

1.	Breed only once in their lifetime	Pacific Salman fish, bamboo
2.	Breed many times	Most birds & Mammals
3.	Produce large number of small sized offspring	Oysters, pelagic fishes
4.	Produce small size of number of large sized offspring	bird, Mammal
5.	Introduced into Australia in early 1920	Prickly pear cactus
6.	Predator	Starfish pisaster
7.	Extinct within a decade after goats were introduced on the island	the Abingdon tortoise
8.	Lice on human & ticks on dog ,Cuscuta, Parasitic plant on hidge plant	Ectoparasite
9.	Various pathogenic bacteria Virus, fungi, protozoan, flat worm, round worm	Endoparasite
10.	Brood Parasitism in birds	Cuckoo (koel) & Crow
11.	Growing on back of whale	Barnacles
12.	Growing as an epiphyte on Mango branch	Orchid
13.	Close association of	Cattle Egret & grazing Cattle
14.	Macarthur warbelers	Resource portioning
15.	Barnacle -gause exclusion	Balanus-superior, cathamalus-inferior
16.	Flamingo and fish compete for	Zooplankton
17.	Pre reproductive and reproductive equal	Stabilising
18.	Pug mark and faecal pellet	Tiger census

Ecosystem

1.	Man made ecosystem	Aquarium, crop land,garden
2.	Vertical distribution of species	Stratification
3.	Unit of productivity	g/m2/yr kcal/m2/yr
4.	Available biomass for next trophic level	NPP
5.	Detritivore rich in lignin and chitin	Fast decomposition
6.	Rich in sugar and nitrogen	Fast decomposition
7.	50 percent of total light	PAR
8.	Inverted pyramid	Aquatic-biomass, forest-number
9.	Aquatic more food chain	GFC
10.	Land more food chain	DFC
11.	Trophic level of secondary carnivore	4^{th}
12.	Black amorphous, colloidal	Humus
13.	Biomass of living at trophic level	Standing crop

Biodiversity and conservation

1.	Robert may	7 million
2.	Genetic level biodiversity	Rauwolfia,Reserpine
3.	Equator more diversity in	Amazon ,Colombia,equador
4.	India area and biodiversity	2.4 percent area ,8.1 percent biodiversity
5.	Maximum biodiversity	23.5 degree S and N
6.	Species area relation ship	AV Humboldt
7.	Long term External plot experiment	Tilman
8.	Extinct from Russia and Africa	Stellar sea cow and Quagga
9.	Value of Z	0.1 to 0.2

Botany

10.	Niche specialization	less seasonal equator
11.	IUCN threatened -most	Amphibian 32 percent
12.	Broadly utilitarian	O2,pollination and aesthetic pleasure
13.	Lake Victoria-invasion by	Nile perch
14.	Over exploitation	Stellar sea cow , passenger pigeon
15.	Habitat loss	amazon forest
16.	Hot spot	25 to 34 (india-3 inod Burma,Himalaya.western ghat
17.	Meghalaya-sacred groove	Garo khasi and Jaintia hills
18.	Wing in rivet popper	keystone species
19.	Ex situ	botanical garden,zoo,wildlife safari park,cryopreservation
20.	Earth summit	Rio-1992
21.	Ramasar site	Wetlands
22.	African cat fish	Foreign species invasion